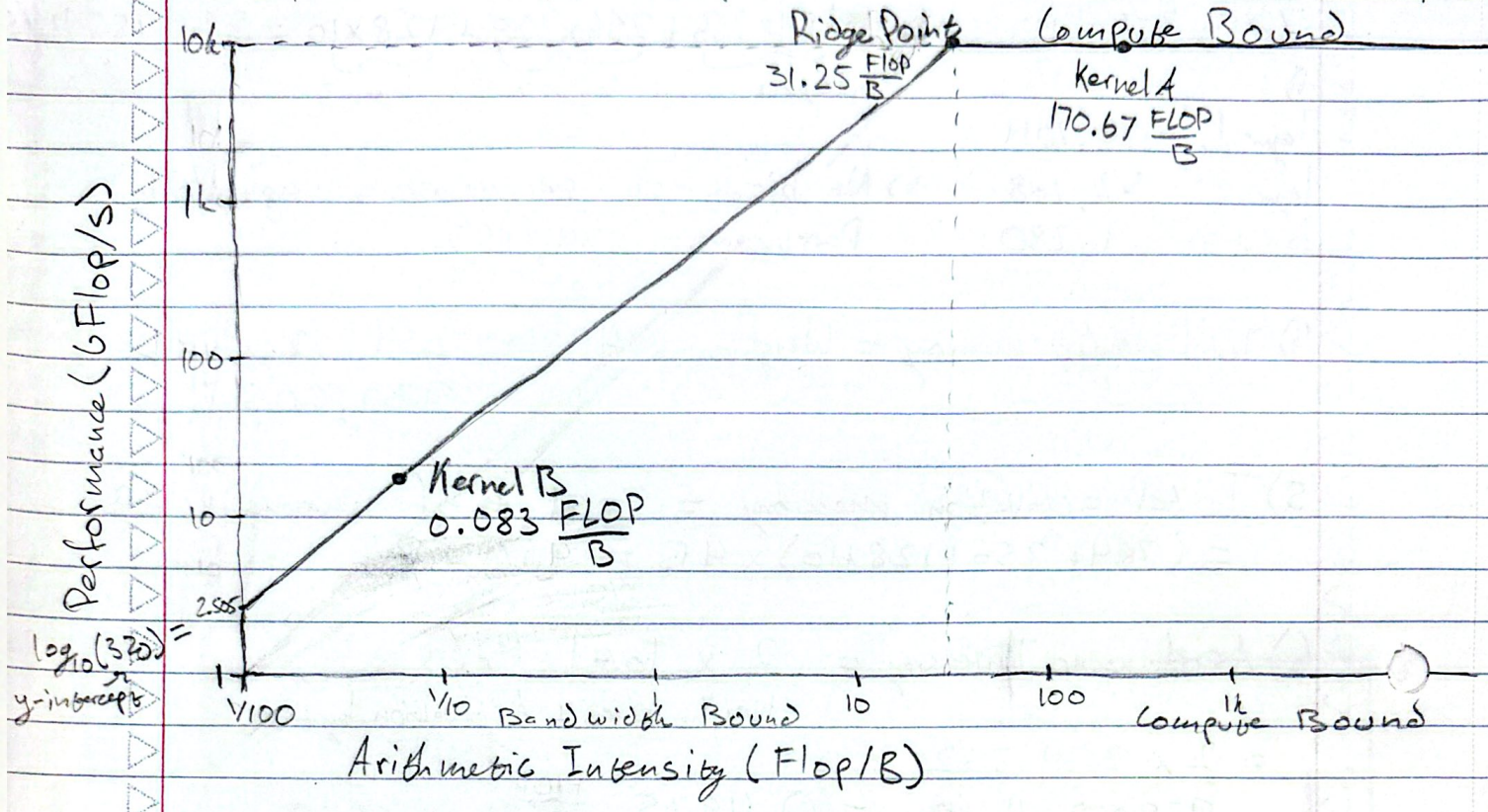
Codefest 2: CMAN: Roofline Construction + Kernel Classification

$P_{peak} = 10$ TFlop/s $B_{peak} = 320$ GB/s RP = 31.25 Flop/B



2) **Kernel A**

- FLOP/s $= 2 \times N^3 = 2G$ FLOPs
- Bytes Transferred $= 3 \times N^2 \times 4 = 12$ MB
- $AI = \frac{\text{FLOPs}}{\text{Bytes}} = \frac{2G \text{ FLOPs}}{12 \text{ MB}} = 170.67 \frac{\text{FLOP}}{\text{B}} = \frac{1024}{6}$

3) **Kernel B**

- FLOP/s $= N = 4{,}194{,}304$
- B Transferred $= 3N \times 4 = 50{,}331{,}648$
- $AI = \frac{\text{FLOPs}}{\text{B}} = \frac{4{,}194{,}304}{50{,}331{,}648} = \frac{1}{12} = 0.083$ FLOP/B

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4) Kernel A

a) Compute Bound

b) Kernel A is compute bound so its performance ceiling is the peak performance of the system or 104 GFLOP/s

c) An architectural change that could improve performance is using a specialized matrix unit like a Tensor core which is faster for systolic array operations

Kernel B

a) Memory Bound

b) Memory Bound so using peak bandwidth of the system.

$$320 \frac{GB}{s} \times 0.083 \frac{FLOP}{B} = 26.67 \frac{GFLOP}{B}$$

c) An architectural change that could improve performance is using wider registers so that more data can be loaded for each instruction.

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